

UNIVERSITY OF OSLO

BedreFlyt Artifact: A Self-Adaptive Digital
Twin Software System for Hospital
Resource Allocation

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ACSOS 2026

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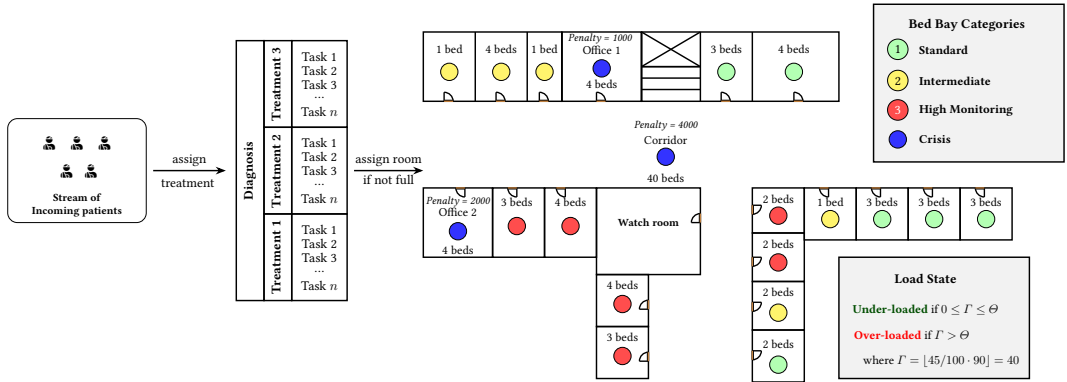
September 10, 2026



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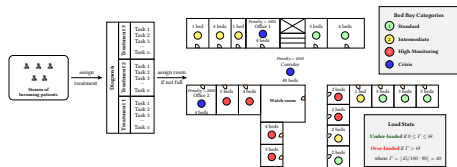
- ① Rationale
- ② Digital Twin Concepts
- ③ Digital Twin Architecture

What do we need?



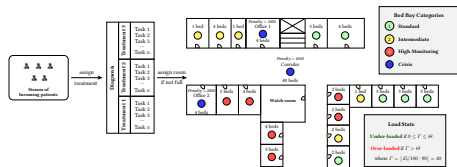
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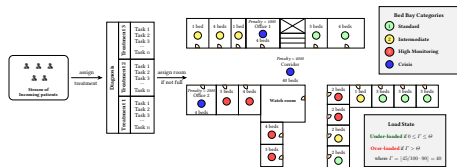
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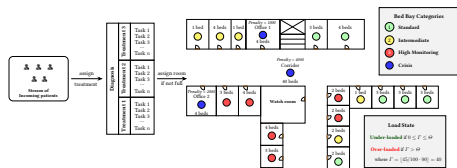
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- Depends on several factors

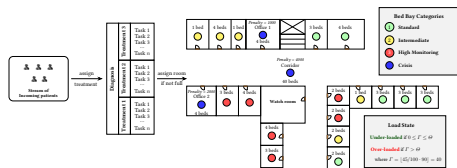


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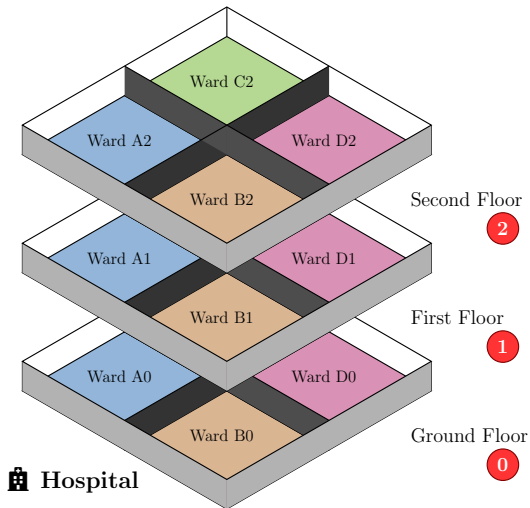


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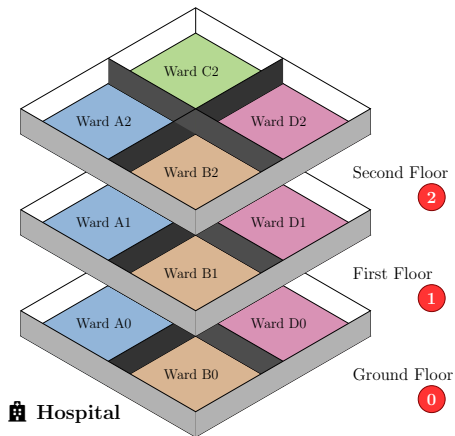


- Hospital demand changes quickly; resource capacity does not
- Admission is often handled manually
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- We want a tool that can help us in the allocation process

What do we need? (cont)

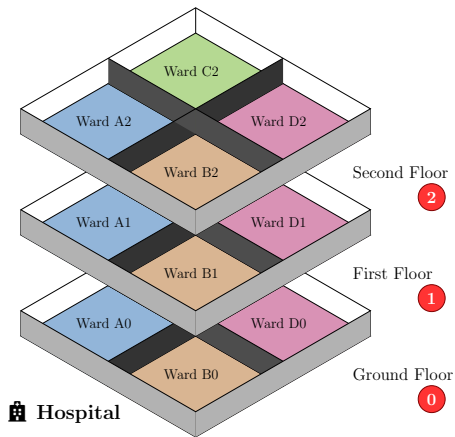


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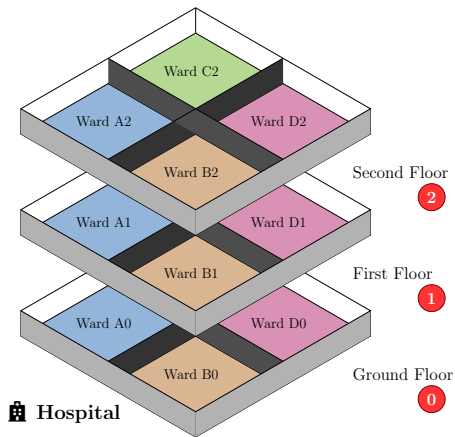
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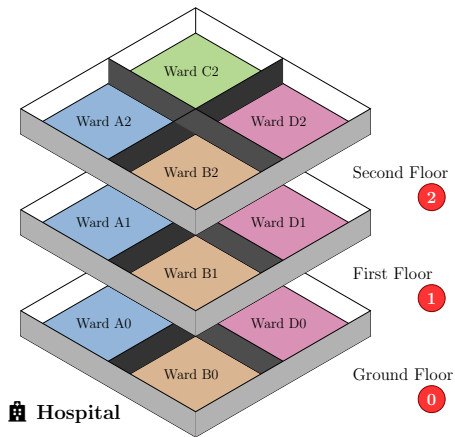
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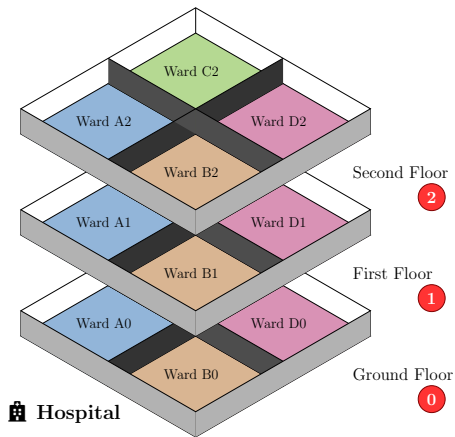
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- If not enough space is free they might be moved
- This requires knowledge of the layout and strategies for interaction
- We want to place the patient as close as possible to the intended ward

Contents

- 1 Rationale
- 2 Digital Twin Concepts
 - Modelling aspects
- 3 Digital Twin Architecture

What is a Digital Twin

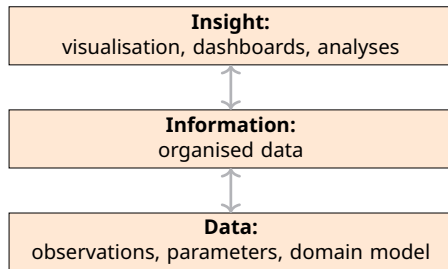
Digital twins are, as defined by NASEM

NASEM Definition of DT (2024)

A digital twin is a set of virtual information constructs that mimics the structure, context, and behaviour of a natural, engineered, or **social system** (or system-of-systems), is dynamically updated with data from its physical twin, has a predictive capability, and **informs decisions** that realise value. The bidirectional interaction between the virtual and the physical is central to the digital twin^a.

^aNational Academies of Sciences, Engineering, and Medicine (NASEM) (2024): Foundational Research Gaps and Future Directions for Digital Twins. The National Academies Press

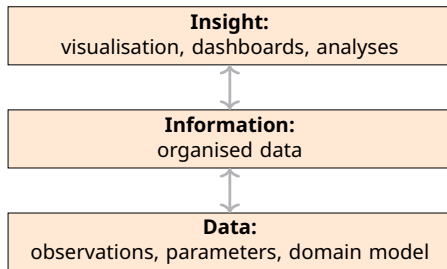
Digital Twins components



Capabilities for different insights:

- **Descriptive:** Insight into the past ("what happened")

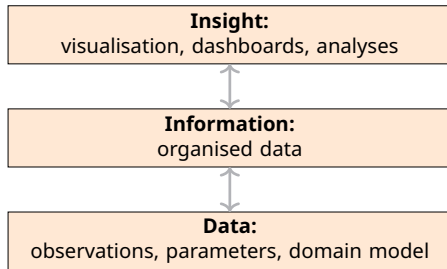
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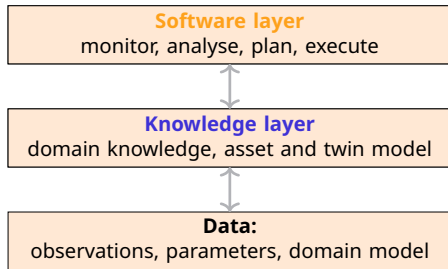
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- **Prescriptive:** Advise on possible outcomes ("what if / what should")

Digital Twins components

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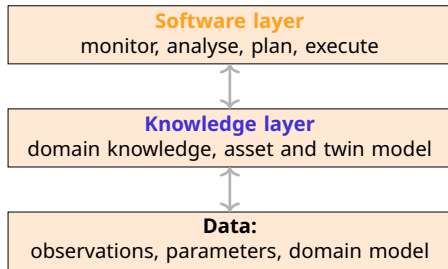
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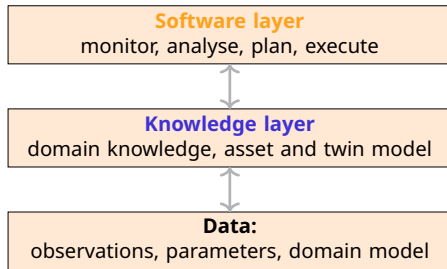


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- DT must adapt both structurally and behaviourally
- States of the DT cycles over time and they need to be captured

Twinning in SMOL

Programming support for twins with a behavioural and a structural layer
smolang.org



Twinning in SMOL

Programming support for twins with a behavioural and a structural layer

SMOL: Semantic Model Object Language

- Smalll OO programming system
- Ontology reasoners allow querying the KB
- Used to create the digital model

smolang.org



Twinning in SMOL

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behavioural twins in SMOL

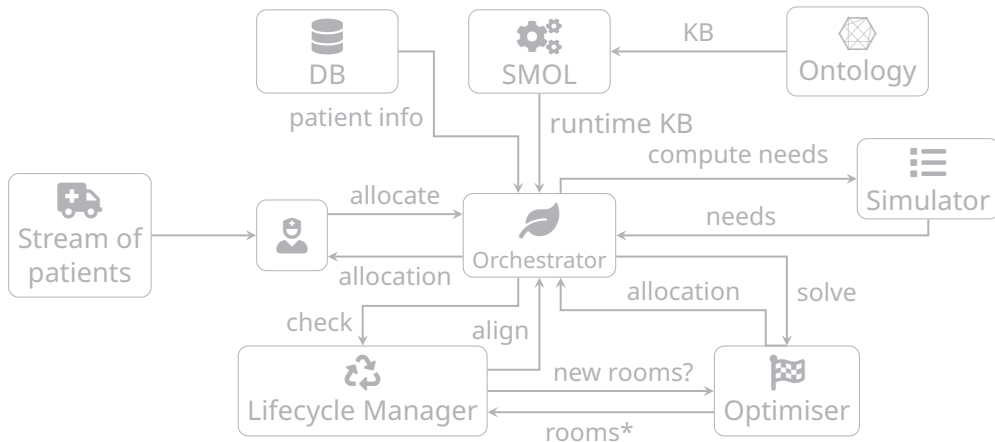
- SMOL can encapsulate (simulation) models based on the FMI standard
- Can automatically adapt the object states in the KB at runtime through semantic lifting



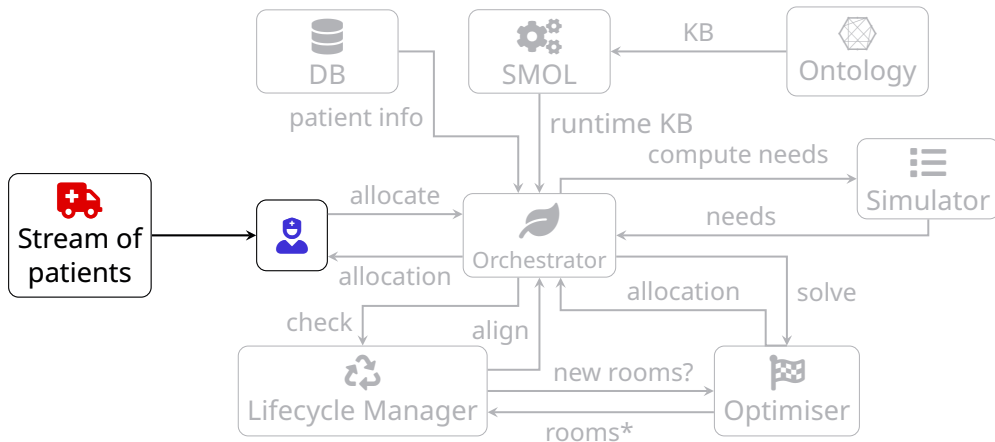
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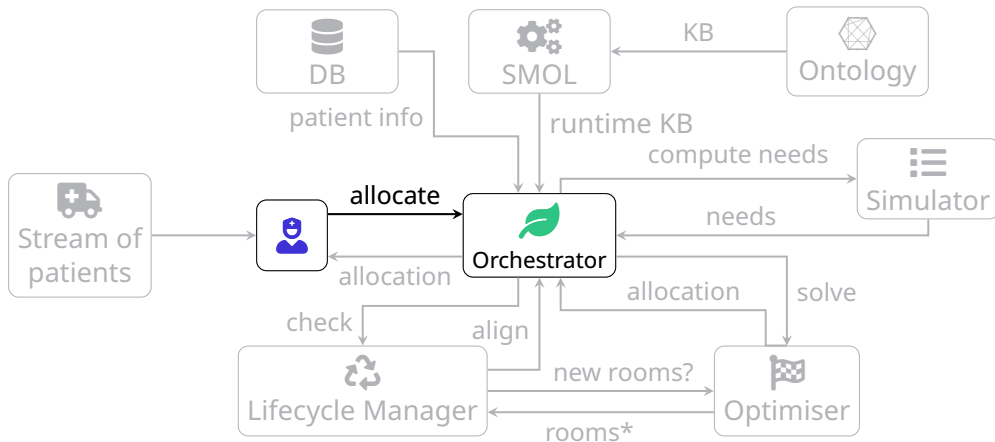
Allocation of patients



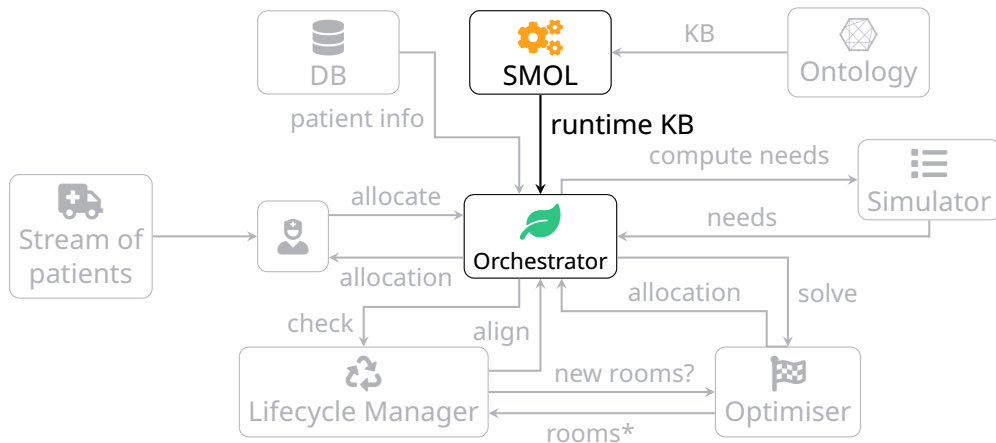
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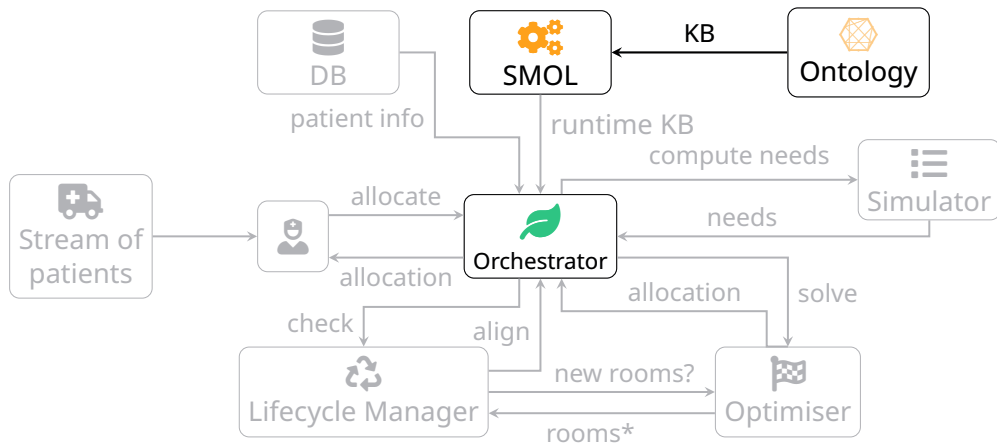
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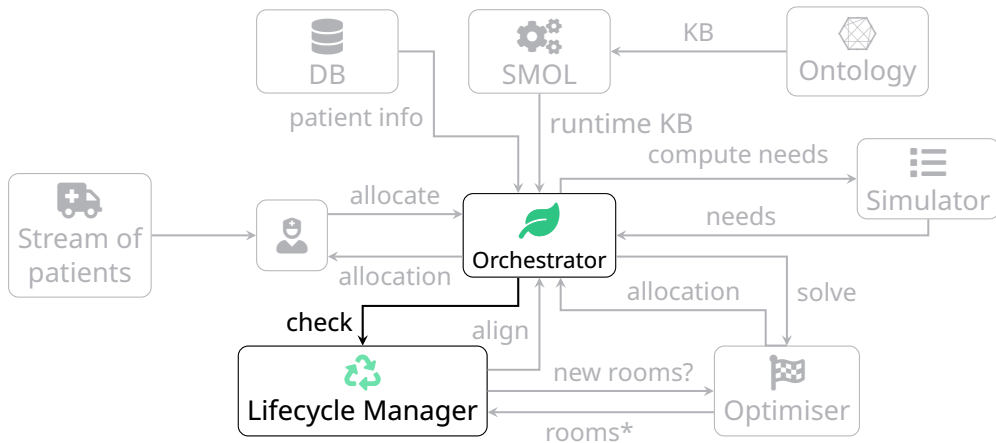
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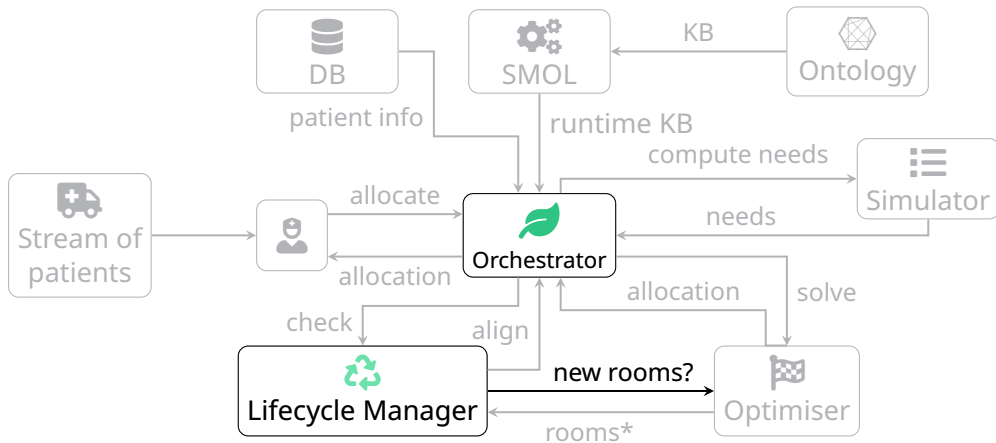
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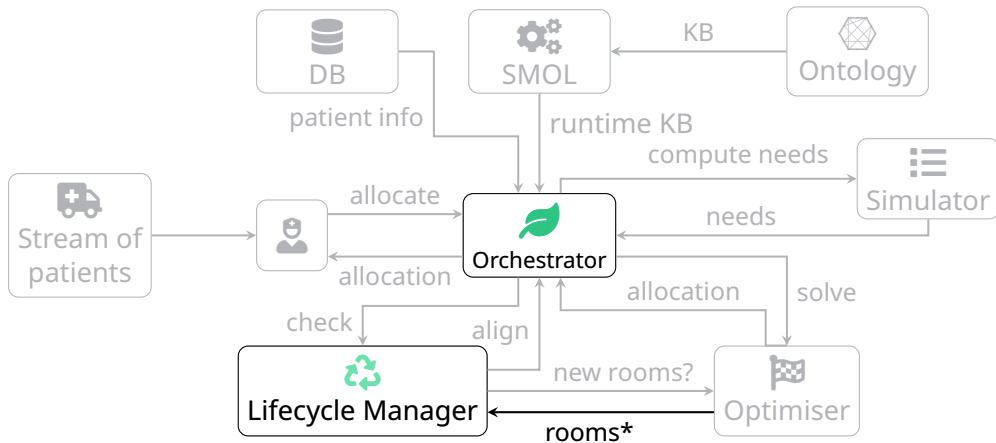
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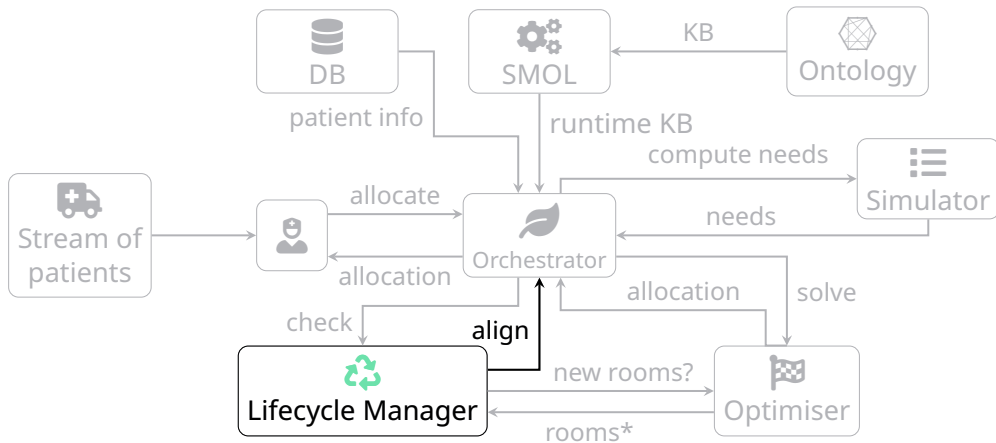
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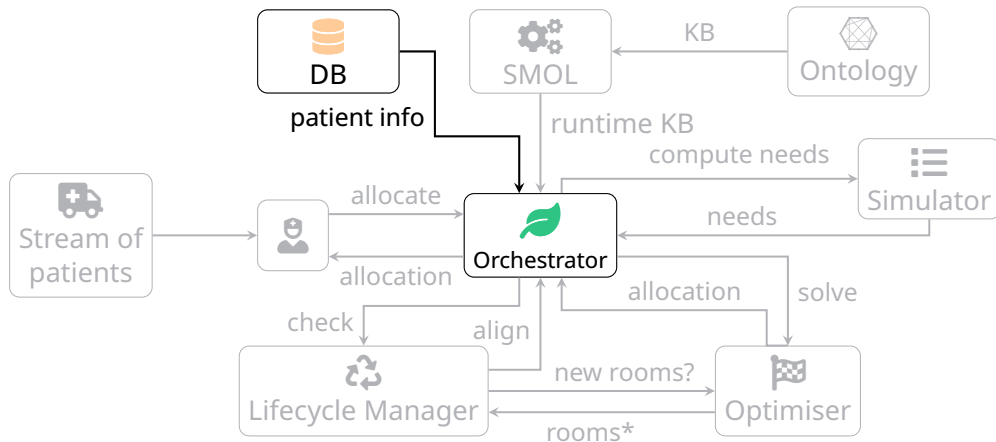
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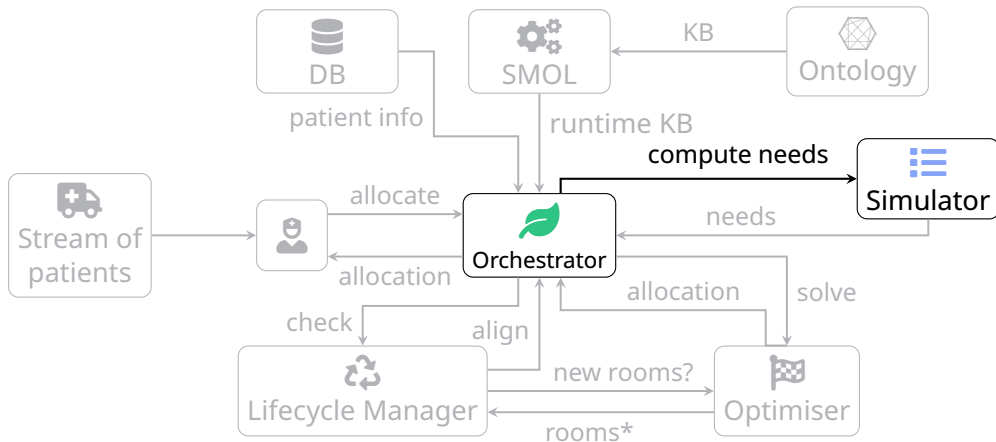
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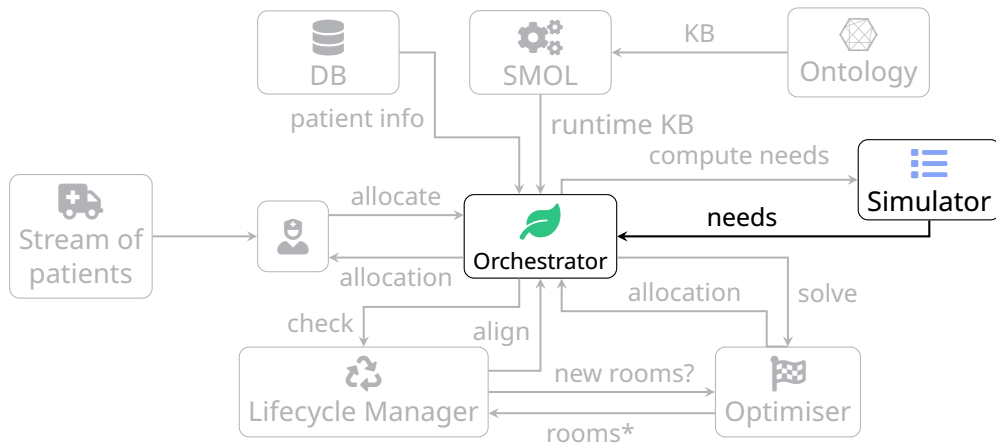
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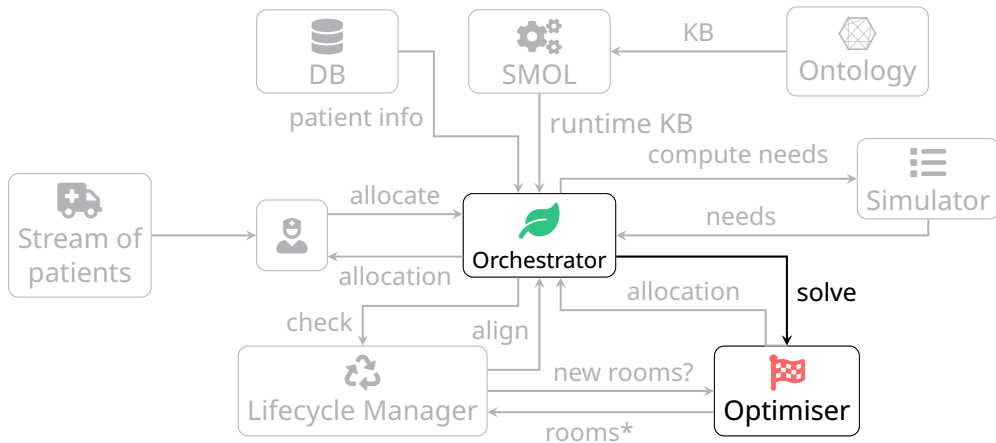
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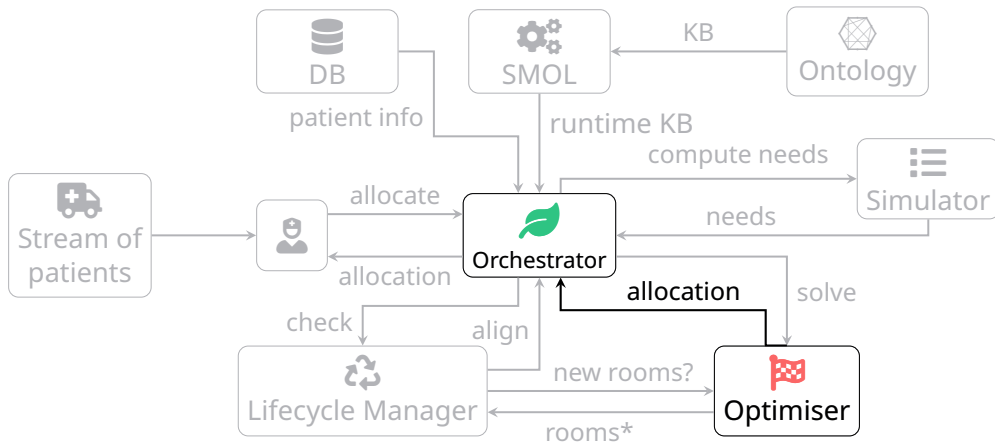
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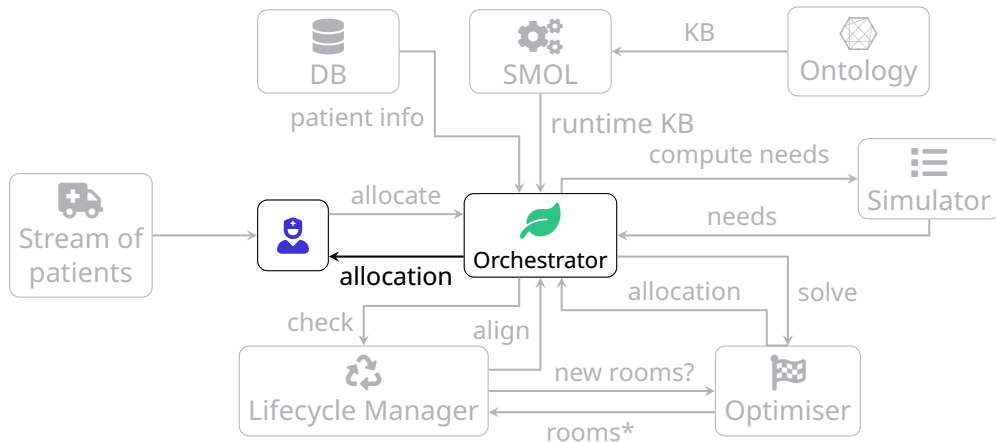
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Human-in-the-loop in the allocation

- The DT proposes the *allocation of patients*

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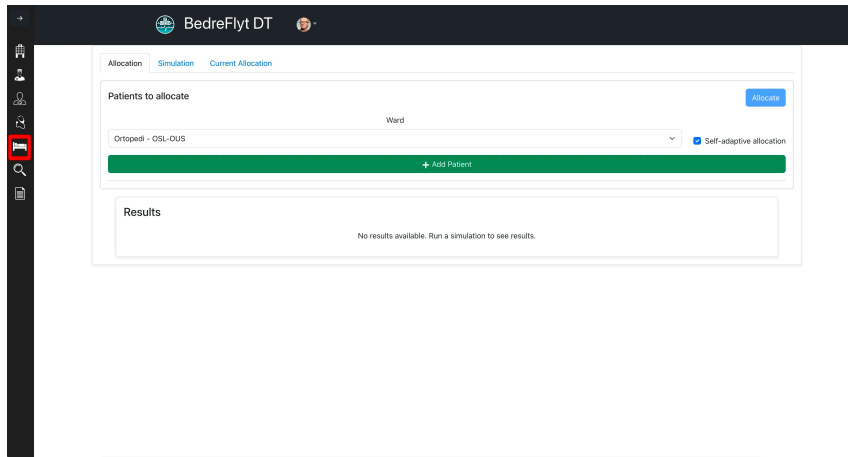
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- The DT proposes the *allocation of patients*
- The human-in-the-loop remains the final authority for adaptation actions
- Approved actions are reflected back via semantic lifting to ensure the model is aligned
- This enables supervised autonomy: controlled self-adaptation instead of static decision support

Run the DT



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- It is also possible to use the Event Stream Generator as a CLI tool simulate stream of patients
- The modularity of the Twin allows the system to change the components as long as the input and output remains consistent
- The DT is being kept updated on GitHub
(<https://github.com/BedreFlyt/bedreflyt>)

Future directions

- **Live data integration** (connect to ward systems rather than simulator surrogate)

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- **Broader scope** (multi-ward, staff constraints, other domains)
- **Live artifact** (make the GitHub repo a live artifact where the new components are merged and the contributors can work on)

Thanks for the attention

Questions?

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